The Embodied Agency Hypothesis: Bioelectric Regulation, Cognitive State, and the Construction of Lived Reality

Daniel Ray Edgar

July 25, 2026

Abstract

Claims about manifestation, bioelectricity, quantum biology, and human divinity circulate in a mixture of accurate physiology, disputed experiments, metaphysical interpretation, and invented precision. The mixture contains a real insight: bodily state changes what an agent can perceive, organize, and reliably bring about. This paper develops the *Embodied Agency Hypothesis* (EAH), a bounded and testable account of that insight. I distinguish ontic reality, lived reality, and enacted future, then define an executable policy set and accessible future set. Intention changes the probabilities assigned to policies; biology constrains which policies can be represented and executed; action transmits the effect into the environment; and feedback updates the next cycle. Under the ordinary causal model, intention has no effect on an external outcome when its behavioral and environmental mediators are fixed.

The synthesis draws on bioelectric morphogenesis, electrophysiology, sleep and circadian disruption, metabolism, hydration, experimental inflammation, ultra-processed diets, the gut–brain axis, notification costs, habits, placebo mechanisms, motor imagery, implementation intentions, and contemplative practice. It also audits claims that words semantically program DNA, the heart has a privileged three-foot field, humans possess a fixed global processing rate, depression is quantum decoherence, and choices select quantum timelines. Molecular quantum effects are real; they do not establish a quantum cognitive computer or voluntary branch selection.

Finally, I treat alchemy, kundalini, Christian theosis, and Christ consciousness as interpretive maps of disciplined transformation rather than substitute anatomy. The result is neither proof that thought creates matter nor denial of spiritual experience. It is a stronger claim: embodied agents construct part of lived reality by changing the distribution of futures that biology, attention, conduct, and environment make reachable. Seven falsifiable predictions, including negative tests, define where the theory ends.

1 Introduction: Recovering the Signal

There is a genre of argument that begins with a familiar biological fact and ends, within a few sentences, with a total theory of reality. DNA is electrically active, therefore it is an antenna. The heart produces a magnetic field, therefore it broadcasts consciousness. Enzymes exhibit quantum tunneling, therefore choice selects a universe. Sleep loss impairs judgment, therefore insomnia has decohered a personal quantum field. The movement from fact to conclusion is fast enough that the missing premises are easy to miss.

These arguments should not be accepted as science. They should not be dismissed without diagnosis either. Their cultural power comes from fastening unsupported mechanisms to an experience that is difficult to deny: a rested, nourished, focused person inhabits a different practical world from an exhausted, inflamed, distracted one. Opportunities are noticed or missed. Impulses are inhibited or enacted. A plan is represented clearly or not at all. The external world has not become arbitrary, yet the set of futures the person can reliably reach may have changed.

This paper asks whether that core intuition can be made precise. I use *manifestation* in a deliberately bounded sense: the process by which an intention, acting through an embodied agent, changes the distribution of future states through attention, policy selection, action, environmental modification, and feedback. This definition is less spectacular than mental control over matter, but it is also more powerful than positive thinking. It specifies a causal system, identifies points of failure, and produces experiments.

Three distinctions organize the argument. First, a physical fact, a scientific model, a metaphor, and a spiritual interpretation are not interchangeable kinds of claim. Second, the electrical character of life does not by itself imply quantum computation or nonlocal agency. Third, *reality* can refer to at least three different things: the world independent of a present experience, the world as perceived and valued by an agent, or the future world produced partly by that agent's actions. Many disagreements about manifestation disappear once the referent is made explicit.

The paper makes four contributions. Section 2 introduces an epistemic ladder for keeping evidence and interpretation in contact without collapsing them. Section 3 audits the scientific premises concerning DNA, cell counts, electromagnetic fields, bioelectricity, neural information, and quantum biology. Section 4 presents the formal EAH model and derives two propositions: the mediator requirement and biological contraction of accessible futures. Sections 5 through 8 connect the model to biological state, goal pursuit, commercial incentive systems, and symbolic traditions. Sections 9 through 11 expose the theory to direct claim-by-claim audit, predictions, counterarguments, and ethical limits.

The aim is not to domesticate every spiritual claim into neuroscience. Nor is it to use mystery as a license for weak inference. The aim is to say exactly what is known, what follows from it, what remains a meaningful interpretation, and what currently has no evidence.

2 An Epistemic Ladder: Fact, Model, Metaphor, and Myth

Claims about mind and body often become confused because four legitimate activities are compressed into one sentence. Measurement reports what an instrument or experiment detected. Modeling proposes a causal or mathematical organization of those measurements.

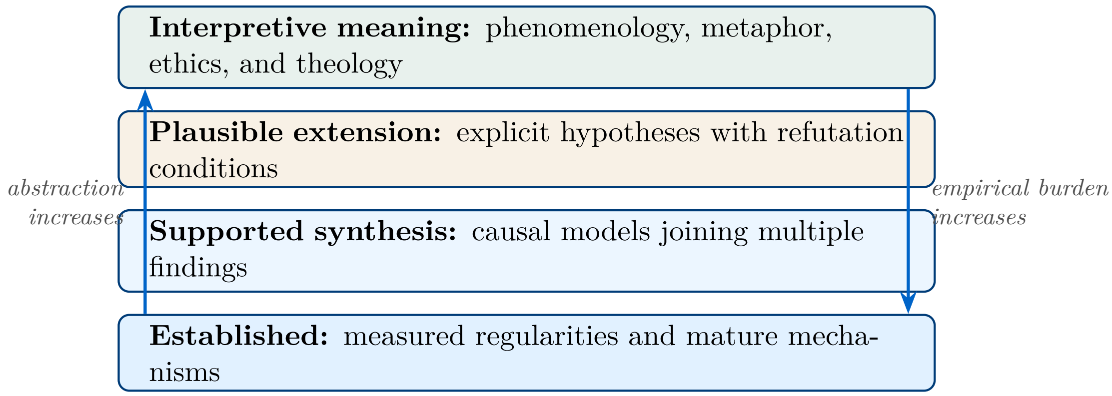

Figure 1: The epistemic ladder. A higher rung is not automatically less valuable, but it cannot be passed off as a lower rung. An interpretation becomes a scientific mechanism only when operational definitions, measurements, and risky tests connect it to the evidence below.

Metaphor transfers structure from one domain to another. Myth, in the technical rather than pejorative sense, organizes identity, value, and ultimate meaning. A proposition can be valuable at one level and false at another.

Definition 2.1 (Evidence status)**.** A claim in this paper is *established* when it rests on mature physical theory or replicated causal evidence; *supported* when convergent evidence exists with material limitations; *plausible* when it is a testable extrapolation not yet established; and *interpretive* when it expresses phenomenological, philosophical, or theological meaning rather than a laboratory finding.

Figure 1 makes two rules visible. First, upward movement requires an argument. Quantum tunneling in an enzyme does not, without several new premises, entail quantum control of life choices. Second, downward movement requires an experiment. If "frequency" is supposed to name a causal biological variable, it must specify what oscillates, at what frequency, with what amplitude, where it is measured, and what intervention changes the outcome.

The ladder also protects spiritual language from bad science. Christian theosis does not become more meaningful if assigned an invented glandular pathway. Kundalini phenomenology does not require every image in a tantric text to name a modern anatomical structure. Alchemical transformation can organize a practice of character without claiming that intention violates conservation laws. Removing a false mechanism does not remove the experience or the ethical demand.

Remark 2.2 (Unknown is not evidence)**.** Failure to possess a complete theory of consciousness does not raise every proposed mechanism to equal probability. Evidence that conscious state covaries with complex brain dynamics [Casali et al., 2013] and that cortical stimulation can alter experienced intention [Desmurget et al., 2009] supports strong brain dependence. It does not settle the metaphysics of mind, but the remaining metaphysical question is not positive evidence for any particular quantum or extracorporeal theory.

3 The Body Is Bioelectric, but Not Thereby a Quantum Computer

3.1 DNA: information, material, and regulation

DNA is both an information-bearing polymer and a physical molecule. The central dogma describes directional transfer of sequence information among nucleic acid and protein systems [Crick, 1970]; it does not say that a genome mechanically determines every future trait. Development depends on gene regulation, cellular context, environmental exposure, stochasticity, and gene–environment interaction [Virolainen et al., 2023]. "DNA is code" and "DNA is not destiny" are therefore compatible statements.

The molecule also has physical properties that the code metaphor does not exhaust. Charge can move through stacked DNA bases under particular conditions, a property used experimentally for electrochemical sensing [Zwang et al., 2018]. This is interesting molecular physics. It is not evidence that ordinary words carry a sequence-independent semantic command that DNA receives and executes.

The modern "linguistic-wave genetics" claim can be traced to papers asserting that speech-like or electromagnetic patterns act on a holographic genome [Gariaev et al., 2011]. The venue is outside the mainstream molecular genetics literature, and a later paper in the same journal states that the reported DNA phantom effect had not, to the author's knowledge, been successfully replicated [Brill, 2012]. A disappearing story is not made stronger by being difficult to find. The scientific burden remains independent replication under blinding, calibrated instrumentation, prespecified analysis, and accessible raw data.

3.2 Numbers that sound more exact than the science

A widely repeated estimate assigns the human body roughly 37 trillion cells. A newer systematic analysis estimates approximately 36 trillion cells in a reference adult male, 28 trillion in a reference adult female, and 17 trillion in a reference ten-year-old child, with body size and counting assumptions materially affecting the answer [Hatton et al., 2023]. The useful conclusion is scale, not a universal personal serial number.

The statement that neurons "fire at 70 millivolts" likewise mixes quantities. Approximately -70 millivolts commonly describes a resting membrane potential, while an action potential is a transient voltage trajectory whose size and shape vary by cell type [Kandel et al., 2021]. Life is electrochemical: concentration gradients, ion channels, membrane capacitance, neurotransmitters, and metabolism form one coupled system. "Not chemistry" is therefore a false contrast.

Nor is there a settled global figure of 400 billion bits per second for a human. Information rate depends on the chosen channel, alphabet, time scale, and decoder. One recent analysis estimated roughly 10 bits per second for selected behavioral outputs [Zheng and Meister, 2025]; a published response argued that this severely understates motor processing and conflates levels of analysis [Sauerbrei and Pruszynski, 2025]. Neither supports a unitary 400-billion-bit meter for consciousness.

3.3 The cardiac field

The heart's electrical currents generate a magnetic field, and magnetocardiography measures it. The relevant signals are commonly on the picotesla scale and require sensitive instruments plus noise control [Roth, 2024]. A magnetic field does not end at a crisp anatomical radius, so detection distance depends on the sensor, geometry, shielding, and background. The popular three-foot figure does not identify a privileged biological boundary. Measurement of a cardiac field also does not by itself demonstrate that the field carries thoughts, coordinates another person's physiology, or constitutes consciousness. Each would require an independent encoding, transmission, and decoding experiment.

3.4 Bioelectric pattern control is real

The strongest scientific version of the "body as an electrical system" thesis lies not in a distant aura but at cell membranes. Membrane potentials and gap-junction networks regulate cell proliferation, migration, gene expression, anatomical patterning, regeneration, and cancer-relevant behavior [Levin, 2021]. In planaria, early bioelectric perturbations can change the polarity of regenerated anatomy, including persistent pattern outcomes after a transient intervention [Durant et al., 2019]. These results expand the causal vocabulary of biology beyond genes alone.

They do not make electrical state omnipotent. Bioelectric signals work through specific cells, channels, transcriptional responses, tissue geometries, and developmental constraints. Their very scientific power comes from those mechanisms. Calling the organism a "bioelectric computer" can be a productive systems metaphor if the states, transitions, memory, and readout are specified. Adding "quantum" does not strengthen the claim unless a quantum computational resource is also identified.

3.5 Quantum biology and the scale problem

All chemistry is quantum mechanical at its foundation. Hydrogen tunneling contributes to specified enzyme reactions [Cha et al., 1989]. Spectroscopic work reported coherence in photosynthetic complexes at physiological temperature [Panitchayangkoon et al., 2010], while later work found no long-lived electronic coherence required for the energy-transfer function [Duan et al., 2017]. This is what a living research program looks like: the relevant state, duration, and function are measured and disputed.

An inference from molecular quantum events to conscious branch selection faces a scale problem. It must show that the relevant states remain coherent long enough, couple to neural computation, survive thermal and environmental noise, influence behavior, and can be controlled by intention. Estimates for proposed neural superpositions have often yielded decoherence times many orders of magnitude shorter than neural dynamics [Tegmark, 2000]. Quantum-consciousness models such as orchestrated objective reduction remain proposals rather than established neural mechanisms [Hameroff and Penrose, 2014].

The many-worlds interpretation is an interpretation of quantum mechanics, not an experimental finding that a person can choose a branch by choosing tea rather than coffee [Vaidman, 2022]. Ordinary decision theory already supplies a sufficient statement: different actions change conditional future probabilities. Invoking literal world splitting adds no predictive leverage unless it yields a result that the ordinary model cannot.

Remark 3.1 (Two meanings of coherence)**.** Quantum coherence is a property of phase relations in a specified quantum state. Psychophysiological coherence is often used more loosely for coordinated function, stable attention, or regulated arousal. The second can be operationalized and useful, but it should not borrow the physical authority of the first. In the rest of this paper, *coordination* names the ordinary biological concept.

4 The Embodied Agency Hypothesis

The scientific content recovered so far is substantial but incomplete. Bioelectric state helps regulate living form; physiological state affects cognition; intention can alter physiology and action. What joins these observations to the construction of reality? The answer depends on which reality is meant.

Definition 4.1 (Three realities)**.** *Ontic reality* is the world that exists independently of an agent's present experience. *Lived reality* is the subset of the world perceived, modeled, valued, and experienced by that agent. An *enacted future* is a later world state whose probability has been changed by the agent's actions.

The distinction is not a retreat into semantics. A belief can radically change lived reality while being false about ontic reality. A repeated action can alter an enacted future without thought directly controlling matter. Conversely, an event can be real while remaining outside the agent's attention and therefore outside the practical world on which that agent acts.

4.1 Executable policies and accessible futures

Let B_t denote the agent's biological state at time t, including relevant neural, metabolic, inflammatory, autonomic, and circadian variables. Let E_t denote the environmental state, I_t an intention, and π a policy, meaning a conditional strategy for action across a time horizon. A policy the agent can imagine but cannot execute is not practically equivalent to one that can be performed reliably.

Definition 4.2 (Executable policy set)**.** For reliability threshold $\rho \in (0, 1]$, define

$$\Pi_\rho(B_t, E_t) = \{\pi : P(\text{execute } \pi \mid B_t, E_t) \geq \rho\}\,. \tag{1}$$

The set includes only policies the agent can both represent and execute above the chosen threshold.

Definition 4.3 (Accessible future set)**.** For outcome threshold $\epsilon \in (0, 1]$, the accessible future set is

$$\mathcal{A}_{\epsilon,\rho}(B_t, E_t) = \{y : \exists\pi \in \Pi_\rho(B_t, E_t) \text{ such that } P(y \mid \mathrm{do}(\pi), B_t, E_t) \geq \epsilon\}\,. \tag{2}$$

This set is not the set of imaginable worlds. It is the set of outcomes reachable with nontrivial probability through at least one executable policy, given present biology and

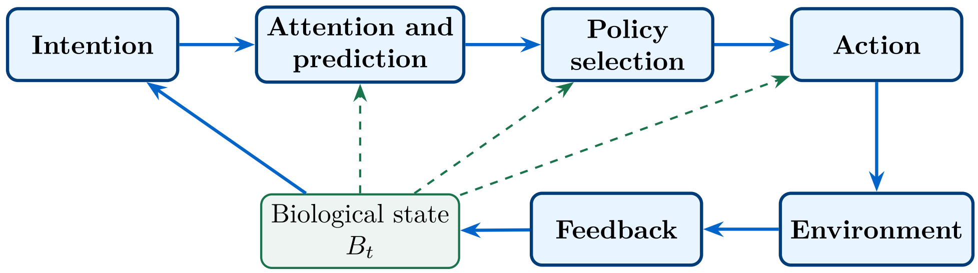

Figure 2: The embodied-agency loop. Biological state does not merely sit upstream of a single choice. It modulates what is noticed, what can be represented, which policy is selected, and how faithfully action is executed. Environmental feedback then changes the next biological and cognitive state.

environment. A future may be possible in principle and inaccessible in practice. The gap between the two is where health, skill, tools, institutions, and social support matter.

Intention changes the distribution of futures by changing policy weights:

$$P(Y \mid I_t, B_t, E_t) = \sum_{\pi \in \Pi_\rho(B_t, E_t)} P(Y \mid \mathrm{do}(\pi), B_t, E_t)\, P(\pi \mid I_t, B_t, E_t). \tag{3}$$

Equation (3) is the *Embodied Agency Operator*, denoted $\mathfrak{E}[I_t; B_t, E_t]$. Intention changes the weights. Biology and environment constrain the support. Action carries the causal effect into the world.

4.2 The mediator requirement

The model makes a sharp commitment about external effects. Consider a causal graph in which intention I affects internal mediators M such as expectation and attention; M affects policy π; and π affects external outcome Y through action. Biology B and environment E condition each stage. Suppose there is no additional direct edge from I to Y.

Proposition 4.4 (Mediator requirement)**.** *Under the causal structure above, fixing the policy and all downstream environmental actions blocks the effect of intention on an external outcome:*

$$P(Y \mid \mathrm{do}(I = i_1), \mathrm{do}(\pi = \bar{\pi}), B, E) = P(Y \mid \mathrm{do}(I = i_2), \mathrm{do}(\pi = \bar{\pi}), B, E) \tag{4}$$

for any intentions i_1, i_2 and fixed policy $\bar{\pi}$.

Proof. Intervention on π removes the structural equation by which upstream intention and its internal mediators select the policy. By assumption, every directed path from I to external Y passes through π or a downstream action fixed by the intervention. The truncated factorization therefore contains no remaining factor through which I changes Y, yielding Equation (4). □

This proposition is a boundary, not a dogma. A reproducible residual effect of intention on a properly isolated external random process would falsify the assumed graph and require

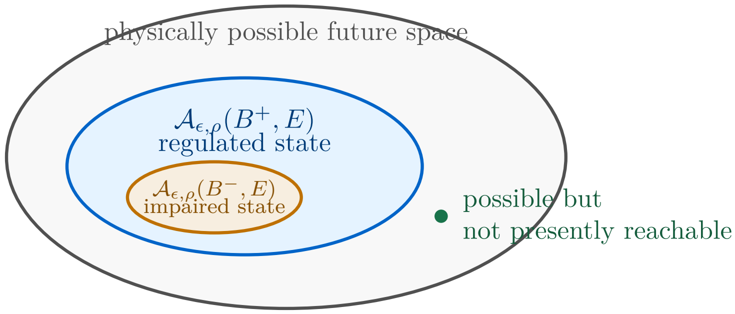

Figure 3: Accessible futures need not equal physically possible futures. An impaired state may contract the policy set, while restoration, skill, tools, or support can expand it. The nested geometry is a hypothesis to be tested for defined tasks, not a claim that all healthy states dominate all impaired states.

a new mechanism. The present literature does not justify building that direct edge into the default model. The burden belongs to experiments that separate intention from expectancy effects, subtle action, sensory leakage, selection, optional stopping, and publication bias.

4.3 Biological contraction

Biology can affect future access even when it does not change the physical possibility of an outcome. Figure 3 illustrates the difference between the full future space and the part accessible above behavioral thresholds.

Proposition 4.5 (Contraction of accessible futures)**.** *Let B^- and B^+ be two biological states in the same environment. If*

$$\Pi_\rho(B^-, E) \subseteq \Pi_\rho(B^+, E) \tag{5}$$

and the outcome distribution for each shared intervention $\mathrm{do}(\pi)$ *is unchanged, then*

$$\mathcal{A}_{\epsilon,\rho}(B^-, E) \subseteq \mathcal{A}_{\epsilon,\rho}(B^+, E). \tag{6}$$

Proof. Take any $y \in \mathcal{A}_{\epsilon,\rho}(B^-, E)$. By Definition 4.3, there exists $\pi \in \Pi_\rho(B^-, E)$ for which $P(y \mid \mathrm{do}(\pi), B^-, E) \geq \epsilon$. The policy-set inclusion places π in $\Pi_\rho(B^+, E)$, and the shared intervention has the same outcome distribution by assumption. Therefore $y \in \mathcal{A}_{\epsilon,\rho}(B^+, E)$. □

The proposition uses deliberately strong assumptions to expose the logic. In actual life, biological state can change not only whether a policy is executable but also the outcome distribution conditional on action. Fatigue may make a negotiation policy less likely to be selected and less effective once attempted. Conversely, states labeled atypical or impaired can expand performance in a particular niche. The empirical question is task-specific set geometry, not a moral ranking of bodies.

Definition 4.6 (Ordinary biological coordination)**.** For a specified task and time scale, biological coordination is the joint adequacy of state regulation, attentional stability, feedback sensitivity, and execution fidelity. It is measured by those components and their behavioral consequences, not by an assumed human quantum field.

5 Biological Bandwidth

The phrase *biological bandwidth* names the resources available for representing, selecting, and executing policies. It is not a literal universal bitrate. The evidence below supports local, task-dependent effects, not a single wellness variable that explains every decision.

5.1 Sleep and circadian alignment

Sleep deprivation does more than produce subjective tiredness. In a controlled study, total sleep deprivation impaired decisions that required participants to update from changing feedback [Whitney et al., 2015]. This maps directly onto the loop in Figure 2: feedback arrives, but policy selection fails to adapt. In a randomized crossover protocol, daily circadian misalignment impaired human cognitive performance in a task-dependent manner [Chellappa et al., 2018]. The task-dependence matters. Poor sleep need not lower every score to alter a consequential choice.

Within EAH, sleep and circadian alignment can affect at least three terms: the size of Π_ρ, the policy-selection distribution $P(\pi \mid I, B, E)$, and the execution reliability embedded in ρ. A person may retain an intention while losing access to the sequence of actions that expresses it.

5.2 Metabolism, hydration, and inflammation

In humans, metabolic state has been experimentally associated with changed economic decisions under risk [Symmonds et al., 2010]. Mild dehydration studies report effects on vigilance, working memory, fatigue, tension, or mood, but the affected measures differ across samples and many cognitive outcomes remain unchanged [Ganio et al., 2011, Armstrong et al., 2012]. The appropriate conclusion is conditional: hydration can influence components of performance under specified conditions, not that a glass of water unlocks a hidden timeline.

Experimental endotoxin provides a more direct test of inflammatory signaling. Dose-dependent immune activation in healthy volunteers changed mood and selected neurobehavioral functions [Grigoleit et al., 2011]. Inflammation is not merely a metaphor for mental static. It is a family of measurable processes that can alter motivation, energy, affect, and cognition. At the same time, no single inflammatory marker explains all depression, and "decoherence" is not a clinical synonym.

5.3 Food and the gut–brain axis

An inpatient randomized controlled trial found that an ultra-processed diet caused greater ad libitum calorie intake and weight gain than an unprocessed diet matched on presented macronutrients, sugar, sodium, and fiber [Hall et al., 2019]. This is strong evidence for a causal dietary effect on intake under the tested conditions. It does not show that every processed ingredient is toxic or that the food supply was coordinated to suppress consciousness.

The gut–brain axis is bidirectional and includes neural, immune, endocrine, and metabolic routes. A small randomized study of healthy participants found that a four-week multispecies probiotic intervention was associated with subtle shifts in microbiome profiles and altered neural responses during emotional tasks [Bagga et al., 2018]. Such evidence makes microbial contribution plausible and sometimes supported. It does not justify the deterministic statement that gut bacteria choose a person's mood, nor does one strain or mixture generalize to the entire microbiome.

5.4 Accumulation without a quantum field

These variables can interact. Sleep changes food choice; diet influences metabolic state; inflammation affects fatigue; fatigue increases default responding; default responding changes the environment encountered tomorrow. A repeated loop can create the subjective experience of being trapped because the same state selects the same policy and the same policy recreates the same state.

This account explains a "narrow band of probability" without positing macroscopic quantum decoherence. The band is the empirically estimated distribution of outcomes under a contracted policy set. Its width can be measured through option generation, behavioral variability, feedback updating, execution fidelity, and actual goal attainment.

6 Manifestation Without Magic

The word *manifest* comes from making something evident or actual. On that meaning, the decisive question is not whether thought matters. It plainly does. The question is which causal path connects a thought to an outcome.

6.1 Expectation changes internal physiology

Expectation can produce measurable biological effects. In Parkinson's disease, placebo expectation was associated with endogenous dopamine release [de la Fuente-Fernández et al., 2001]. In experimental pain, placebo conditions altered both anticipatory and pain-related brain activity [Wager et al., 2004]. These findings are neither "merely imaginary" nor evidence that belief can cause any desired external event. They show that information and expectation enter a biological control system and change its response.

This is an important correction to a simple matter-versus-mind picture. A symbol, a clinician's words, or a learned context can become biologically causal because the organism interprets it. Meaning is not outside physiology once perception, prediction, autonomic response, endocrine signaling, and action are part of the mechanism.

6.2 Imagery is rehearsal, not delivery

Kinesthetic motor imagery can train components of a motor program. A small controlled study reported increased cortical signal and strength following imagery training of forceful contractions [Yao et al., 2013]. The result is compatible with neural rehearsal and altered recruitment. It does not imply that imagining a stronger muscle adds mass without metabolism or that visualizing a contract causes another person to sign it.

Mental simulation is most useful when it improves a subsequent policy. It can expose a missing step, lower novelty, precommit a response, or sharpen cues for action. It can also become a substitute reward that reduces action. EAH therefore predicts effects from the change in policy distribution, not from vividness alone.

6.3 Intention needs an executable grammar

Implementation intentions translate a goal into a conditional rule of the form "if situation X occurs, I will perform action Y." Experimental evidence shows that these rules can improve goal attainment, especially when the underlying goal intention is active and strong [Sheeran et al., 2005]. In the language of Equation (3), the rule raises the probability of a target policy when its cue appears.

The difference between aspiration and implementation is the difference between a desired outcome and an executable grammar. "I intend to be healthy" assigns value. "After I put the coffee on, I will take the prepared medication" binds a cue to an action. Repeated successful execution then modifies the environment and produces evidence that updates self-prediction.

6.4 Practice changes the practitioner

Meditation provides a useful case because spiritual and biological descriptions can be studied together. Experienced meditators have shown differences in default-mode network activity and connectivity during meditation [Brewer et al., 2011]. A small randomized trial in stressed family caregivers reported differences in leukocyte transcriptional responses after a yogic meditation program compared with relaxing music [Black et al., 2013]. Frameworks of mindfulness connect self-awareness, self-regulation, and self-transcendence to trainable cognitive and neural processes [Vago and Silbersweig, 2012].

These findings warrant cautious support for state and trait change. They do not establish a single frequency of enlightenment, and associations in experienced practitioners do not by themselves prove causality. Practice becomes an embodied technology only to the extent that dose, instruction, adherence, comparison condition, outcome, and adverse effects are measured.

6.5 What "frequency" can responsibly mean

Frequency has precise meanings when a periodic process is specified: neural oscillation, respiration, heart rate, acoustic pressure, electromagnetic radiation, or behavioral repetition. It can also function metaphorically as a stable pattern of affect and conduct. Problems arise when the meanings are exchanged mid-argument.

Within EAH, the most defensible version of "you are the frequency" is behavioral: what a person repeatedly attends to and enacts becomes more probable through learning, habit, environment design, and social feedback. This is not a cosmic carrier wave. It is a temporal pattern in a causal system.

Remark 6.1 (A bounded formula for manifestation)**.** A concise statement of the paper is

$$\text{manifestation} = \text{intention} \times \text{embodied capacity} \times \text{policy} \times \text{action} \times \text{feedback}, \tag{7}$$

where multiplication indicates a series system: a near-zero term can constrain the whole pathway. This is a conceptual decomposition, not a calibrated physical law.

7 Manufactured Interference Without a Master Plot

The source thesis attributes processed food, notifications, outrage, sexual degeneracy, and nine-to-five work to a world designed to make people forget their divinity. The evidence supports a narrower structural argument. Systems optimized for consumption and engagement can degrade conditions for reflective agency even when no actor intends spiritual suppression.

7.1 Attention is a rival resource

Receiving a phone notification can disrupt performance even when the recipient does not interact with the device [Stothart et al., 2015]. Behavioral and electrophysiological work likewise finds costs to cognitive control from smartphone notifications [Upshaw et al., 2022]. A single interruption is small. A system of repeated cues can repeatedly reset the competition for attention, fragment long-horizon policies, and increase selection of immediately available actions.

Habit research shows why context matters. Stable cues can trigger practiced behavior with limited deliberation, while a context change can disrupt the habit and return behavior to more intentional control [Wood et al., 2005]. "Default settings" are therefore not merely rhetoric. They can be described as learned cue-policy mappings whose selection probability has become insensitive to a current high-level goal.

7.2 Optimization produces externalities

Industrial recommendation systems are explicitly built to rank items from enormous candidate spaces using behavioral data and objectives related to engagement and watching [Covington et al., 2016]. Food products compete for repeat purchase and palatability; an inpatient trial shows that an ultra-processed dietary environment can increase intake under controlled conditions [Hall et al., 2019]. These are documented objective functions and outcomes.

No secret council is required. If platforms profit from attention, content that holds attention receives systematic advantage. If producers profit from consumption, products that encourage consumption receive systematic advantage. The aggregate environment can become hostile to sustained agency through local optimization. Calling this a conspiracy weakens the critique because it replaces measurable incentives with an unnecessary hidden intention.

7.3 Moral language and scientific language

"Degeneracy" is a moral and political judgment, not an operational psychological variable. A professional account must specify the proposed harm: compulsivity, exploitation, loss of consent, attentional capture, sleep displacement, isolation, or conflict with a person's chosen commitments. This permits measurement and protects against labeling unfamiliar identities or harmless pleasure as pathology.

EAH assigns causal responsibility at more than one level. Individuals can redesign cues and practice policies. Firms select objective functions. Institutions regulate externalities. Social conditions change the cost of rest, food, safety, and contemplation. "Treat your body well" is incomplete when environments constrain the available policies; "the system did it" is incomplete when people can still create local structure. Agency is embodied and situated.

8 Ancient Symbolic Maps of Transformation

The traditions grouped together as alchemy, kundalini, sacred secretion, and Christ consciousness are historically distinct. Treating them as one secret physiological manual risks erasing their differences. Yet comparison can be illuminating when it is offered as interpretation rather than proof of a single hidden anatomy.

8.1 Alchemy: matter, symbol, and disciplined transformation

Historical alchemy included practical work with substances, cosmology, medicine, and spiritual aspiration. The Hermetic corpus links knowledge of cosmos, mind, and divine order [Copenhaver, 1992]; South Asian alchemical traditions developed their own embodied and soteriological systems [White, 1996]. In Jung's psychological reading, alchemical images become symbols of integration and transformation [Jung, 1968].

EAH offers a limited scientific analogue. Repeated practice can transform the agent who perceives and acts, and that altered agent changes future outcomes. The *prima materia* in this analogy is not a supernatural fluid but the coupled state of attention, body, habit, and environment. The analogy is useful because it emphasizes work, sequence, and feedback. It ceases to be science if each symbolic operation is declared a literal biochemical event without measurement.

8.2 Kundalini: text and phenomenology

Kundalini belongs to particular tantric and later yogic textual histories. The *Saradatilakatantra*, for example, presents a structured yoga involving centers, breath, visualization, and a coiled power [Bühnemann, 2011]. Modern interview research records spiritually transformative experiences described as kundalini awakening, including somatic, affective, perceptual, and life-changing features [Woollacott et al., 2021]. Such interviews establish reported phenomenology, not a spinal energy substance.

A rigorous research program could measure breathing, autonomic function, sleep, movement, endocrine state, neural activity, expectation, cultural framing, and long-term outcomes during defined practices. It should also measure distress and impairment. The categories "real experience" and "established anatomical mechanism" must not be collapsed.

8.3 Theosis and Christ consciousness

Christian theosis names participation in divine life, classically expressed in Athanasius's account of incarnation and developed across Christian theology [Athanasius of Alexandria, 2011, Papanikolaou, 2020]. It concerns transformation of the person in relation to God, virtue, communion, and grace. "Christ consciousness" is a later and broader phrase, often used for a mode of unity, love, self-giving, or divine awareness. It is not a standard neuroscientific construct.

The empirical and theological statements can coexist without pretending to be the same statement. Contemplative discipline may train attention and regulation. Self-transcendent experiences form a recognizable but heterogeneous psychological family [Yaden et al., 2017]. Theology interprets the source, purpose, and ultimate meaning of transformation. Neuroscience measures correlates and consequences. Neither discipline gains by disguising its question as the other's answer.

8.4 The sacred-secretion claim

Some modern syntheses propose a monthly "sacred secretion" that descends from the brain, rests near the sacrum, rises through the spine, and becomes Christ oil when preserved. No established anatomical literature demonstrates that cycle. Cerebrospinal fluid is continuously produced, circulated, exchanged, and cleared through physiological pathways more complex than the proposed monthly descent and ascent [Hladky and Barrand, 2014].

Endogenous *N,N*-dimethyltryptamine has been measured in mammalian systems, but rat-brain work indicates that its synthesis is not confined to the pineal gland [Dean et al., 2019]. Reviews specifically caution against turning pineal DMT speculation into fact [Nichols, 2018]. Neither finding establishes the sacred-secretion cycle.

The symbolic comparison can still carry meaning: disciplined attention, restraint, prayer, breath, and ethical alignment are described as preparing a person for transformation. That is an interpretive thesis. Its dignity does not depend on calling cerebrospinal fluid an oil or assigning Christ to a molecule.

Remark 8.1 (Plural maps, partial overlap)**.** Alchemy emphasizes transformation, kundalini traditions articulate embodied ascent, and theosis describes participation in divine life. Their partial structural overlap does not prove common origin or identical mechanism. Comparison should preserve both resonance and difference.

9 Claim Audit

Table 1 applies the epistemic ladder to the most specific claims in the source thesis. A mixed verdict does not mean that every part is half true. It means an accurate premise has been joined to a conclusion that needs separate evidence.

Table 1: Audit of the source thesis.

Claim	**Verdict**	**Evidence and boundary**
Russian researchers proved that words and frequencies reprogram DNA.	**Unsupported**	The linguistic-wave claim is documented [Gariaev et al., 2011], but the reported phantom effect lacks successful independent replication [Brill, 2012]. No blinded semantic-programming result is established.

Continued on next page

Table 1 continued

Claim	Verdict	Evidence and boundary
DNA is not code; it is a crystalline antenna.	**False dichotomy**	DNA carries sequence information [Crick, 1970] and has physical charge-transport properties [Zwang et al., 2018]. Neither implies reception of semantic commands, and physical behavior does not cancel genetic information.
The heart broadcasts a measurable field three feet around the body.	**Partly true**	Magnetocardiography measures picotesla-scale cardiac fields [Roth, 2024]. A field has no crisp three-foot edge, and interpersonal transmission of thoughts has not been shown.
Every person has 37 trillion cells, and neurons fire at 70 millivolts.	**Misleading precision**	Cell estimates vary with body and method [Hatton et al., 2023]. Approximately −70 mV usually names resting potential, not a universal firing value [Kandel et al., 2021].
Humans process 400 billion bits per second.	**Unsupported**	There is no single agreed channel called whole-human processing. Even a recent 10-bits-per-second behavioral estimate [Zheng and Meister, 2025] has been directly challenged [Sauerbrei and Pruszynski, 2025].
The body is a bioelectric quantum computer.	**Mixed**	Bioelectric networks regulate tissues [Levin, 2021]; molecular quantum effects occur [Cha et al., 1989]. Evidence for quantum computation supporting human cognition remains absent, with serious decoherence constraints [Tegmark, 2000].
Each choice selects which quantum timeline is experienced.	**Interpretive**	Different choices alter ordinary conditional futures. Many-worlds does not provide a demonstrated mechanism for voluntary branch selection [Vaidman, 2022].
Sleep loss, inflammatory food, and ambient EMF decohere the human field.	**Mixed and undefined**	Sleep, circadian state, diet, and experimental inflammation affect selected outcomes [Whitney et al., 2015, Chellappa et al., 2018, Hall et al., 2019, Grigoleit et al., 2011]. No measured personal quantum field or decoherence variable connects them. Exposure claims require specified frequency and dose [International Commission on Non-Ionizing Radiation Protection, 2020].
Gut bacteria and dehydration determine mood and decision-making.	**Overstated**	Human intervention evidence supports some state effects, with small samples, strain-specificity, task-dependence, and null outcomes [Bagga et al., 2018, Ganio et al., 2011, Armstrong et al., 2012]. Influence is not determination.

Continued on next page

Table 1 continued

Claim	Verdict	Evidence and boundary
Depression is decoherence.	**False category**	Depression is a heterogeneous clinical syndrome. Quantum decoherence is a physical process in a specified state. Replacing diagnosis with metaphor can delay appropriate assessment and care.
The world was designed to make people forget their divinity.	**Unsupported as coordination**	Notifications disrupt attention [Stothart et al., 2015], habits follow context [Wood et al., 2005], and recommendation systems optimize behavioral objectives [Covington et al., 2016]. These mechanisms can produce interference without a master plan.
Kundalini, Christ consciousness, and a monthly sacred secretion are one physiology.	**Interpretive overlap; unsupported anatomy**	Tantric texts, Christian theosis, and modern reports describe genuine traditions and experiences [Bühnemann, 2011, Papanikolaou, 2020, Woollacott et al., 2021]. Known cerebrospinal-fluid and pineal evidence does not establish the proposed cycle [Hladky and Barrand, 2014, Nichols, 2018].

The audit reveals a recurring structure. The source material begins with a true statement, removes its scale and conditions, attaches a metaphysical conclusion, and then returns the authority of the conclusion to the original fact. The correction is not to deny the fact. It is to preserve the missing conditions.

10 Falsifiable Predictions and Proposed Tests

A framework that explains every result after the fact explains nothing. The following predictions commit EAH to observable outcomes and, equally important, to negative results.

Prediction 1 (State-dependent expansion)**.** In participants with verified sleep or circadian disruption, restoration of sleep opportunity and timing will increase the number of task-relevant policies generated, feedback-sensitive policy switching, and execution reliability relative to an attention-matched control. Improvements should be mediated by measured state changes, not by a generic belief score alone.

A preregistered study could repeatedly sample sleep, circadian phase, inflammation, metabolic measures, option generation, reversal learning, and real-world execution. The prediction fails in its proposed domain if state restoration changes subjective wellness but produces no reliable change in the policy or outcome measures.

Prediction 2 (The action mediator)**.** Implementation intentions plus environmental cue design will outperform outcome visualization alone on independently verified goal completion. When action frequency and quality are held fixed, visualization will not add a reliable effect on external outcomes that participants cannot otherwise influence.

This is a direct test of Proposition 4.4. The first clause expects a behavioral effect consistent with implementation-intention research [Sheeran et al., 2005]. The second is the negative boundary. A persistent, multilaboratory residual effect on isolated external outcomes would falsify the ordinary causal graph.

Prediction 3 (Context reopens control)**.** Removing high-frequency notification cues and changing the physical context of a target habit will increase goal-consistent choice most strongly for participants whose baseline behavior is cue-driven. The effect will be mediated by fewer attention switches and reduced automatic policy selection.

The prediction is stronger than "phones are distracting." It specifies a moderator, a mediator, and an outcome. Failure of the measured mediators to account for the behavioral change would require revision of the mechanism.

Prediction 4 (No unitary biological bitrate)**.** Estimates of human information rate will differ by orders of magnitude when computed for sensory transduction, neural population activity, motor control, conscious report, and task-level behavior. No context-free scalar will predict performance across all levels better than level-specific measures.

This prediction can be rejected by a well-defined common coding model whose single rate generalizes across levels and tasks. Until then, very large or very small global bitrate claims should be treated as properties of the measurement choice.

Prediction 5 (Semantic DNA programming fails)**.** In a blinded, multilaboratory experiment with calibrated physical exposure, speech samples matched for acoustic energy but differing in semantic meaning will not produce sequence-independent, meaning-specific changes in purified DNA or cells beyond chance, contamination, and ordinary stress responses.

The refuting result would be a reproducible semantic effect that survives language translation, speaker change, sham exposure, multiple-comparison correction, raw-data release, and mechanistic follow-up. This test allows the extraordinary claim to win, but only by doing experimental work.

Prediction 6 (No sacred-secretion cycle)**.** Intensive longitudinal measurement will not reveal a universal monthly substance that travels from brain to sacrum and back in the pathway asserted by the sacred-secretion account. Reported spiritual intensity may covary with sleep, expectation, practice, autonomic state, hormones, or other measured variables without establishing that anatomy.

This negative prediction separates experience from mechanism. Discovery of a previously unknown reproducible cycle with a defined molecule, route, timing, and causal intervention would falsify it.

Prediction 7 (Practice changes reachable futures through measured mediators)**.** Well-specified contemplative practice will change selected attention, self-regulation, and self-transcendence measures for some populations, and these changes will predict later behavior through measurable mediators. The same practice will not reliably influence shielded random physical outcomes when behavior and communication are excluded.

The prediction honors both sides of the epistemic ladder. It expects embodied transformation and rejects the need to inflate that transformation into unbounded physical control.

Taken together, Predictions 1 through 7 define a research program. The central dependent variable is not a mood adjective called "high frequency." It is the geometry of accessible action and outcome: how many relevant policies are represented, how reliably they are executed, how quickly they adapt to feedback, and which independently verified futures follow.

11 Counterarguments, Limits, and Ethical Cautions

11.1 Is the thesis merely that health affects decisions?

The component facts are not new, and they should not be presented as a secret manual suppressed since the 1990s. The proposed contribution is their organization around executable policies and accessible futures. EAH distinguishes three mechanisms that a generic wellness claim blends together: state can change the policies represented, their selection weights, and their execution reliability. It also commits to the mediator requirement, which blocks an easy slide from intention to unexplained external causation.

This contribution may still prove too coarse. Different biological variables may not share a useful common geometry, and the policy-set metaphor may add little beyond existing cognitive-control and decision models. Predictions 1 and 3 are designed to expose that possibility.

11.2 Does the model reduce spiritual life to physiology?

No physiological measure settles whether God exists, whether consciousness is fundamental, or what a transformative experience ultimately means. EAH is a model of how embodied intention changes lived and enacted reality. It is silent about ultimate ontology beyond requiring that proposed physical mechanisms face physical tests.

This restraint cuts both ways. A neural correlate does not explain away prayer, love, or theosis. A sacred interpretation does not establish a glandular pathway. First- person meaning, communal tradition, ethics, and third-person measurement answer different questions that can constrain one another without becoming identical.

11.3 Does biology become destiny again?

An accessible future set is indexed to a time, environment, threshold, and horizon. It is not a permanent essence. Tools, training, medicine, relationships, accommodations, institutions, and chance can change both B and E. The model is therefore anti-deterministic in the practical sense: it identifies intervention points.

It also rejects the opposite cruelty, the idea that anyone can obtain any outcome by holding the correct thought. Bodies differ. Resources differ. Other agents act. Randomness and structural constraints remain. Failure is not evidence of deficient faith or low vibration.

11.4 Clinical and public-health boundaries

The statement “you are not depressed, you are decoherent” is scientifically false and potentially harmful. Depression can involve urgent risk and deserves qualified assessment. Sleep, nutrition, movement, social connection, contemplative practice, and environment may be relevant to care, but a conceptual framework is not a diagnosis and this paper is not medical advice. Biological state should widen the care model, not become a reason to abandon evidence-based treatment.

Environmental health claims require equal precision. Electromagnetic exposure has frequency, intensity, duration, geometry, and tissue-coupling parameters. Safety guidelines for radiofrequency exposure are written in those terms [International Commission on Non-Ionizing Radiation Protection, 2020]. “EMF saturation” without them is not a testable dose. This does not mean every possible exposure question is closed. It means the question must be specified before fear can count as evidence.

11.5 Measurement can become another cage

An optimization system for sleep, diet, heart rate, productivity, and attention can itself consume the attention it was meant to restore. Biological bandwidth is not a moral score, and no wearable measures consciousness or divinity. The goal is not perfect control. It is sufficient coordination to recover reflective choice and act in accordance with chosen values.

12 Conclusion: Becoming the Broadcast

The human body is electrically organized, chemically sustained, environmentally coupled, and capable of changing itself through learning and practice. DNA carries information without dictating destiny. Cells use membrane potentials to coordinate form. Hearts generate measurable magnetic fields. Expectations alter physiology. Sleep, circadian timing, metabolism, inflammation, hydration, food environments, microbial systems, habits, and notifications can change parts of perception and action. None of these facts requires a hidden quantum manual, and none is made less remarkable by being mechanistic.

The phrase “you create your reality” becomes defensible when reality is specified. An agent constructs lived reality by selecting what is attended to, modeled, and valued. An agent helps construct enacted reality by choosing and repeating actions that alter the environment. Biological state changes which policies can be represented and reliably executed, so it changes the distribution of reachable futures. That is the Embodied Agency Hypothesis.

The theory draws a bright line at ontic control. Quantum tunneling in a cell is not evidence that intention selects a universe. A cardiac field is not evidence of thought transmission. Charge movement in DNA is not semantic programming. A spiritual experience is not proof of a monthly cerebrospinal oil. These claims may be tested, but they cannot be installed as premises before the result.

Alchemy, kundalini, theosis, and Christ consciousness can still illuminate the phenomenology and ethics of transformation. Their shared invitation is not merely to hold an idea but to become the kind of person through whom the idea acquires form. Scientific language translates part of that process into state, policy, action, and feedback. Spiritual language asks what the transformation is for.

To stop being only a signal and become a broadcast, in the strongest defensible sense, is to cease living exclusively as the output of inherited cues. It is to regulate the conditions of attention, choose an executable policy, act, read the consequence, and repeat. The mind does not need exemption from causality to be powerful. Its power is that, through a living body, it enters causality.

References

Lawrence E. Armstrong, Matthew S. Ganio, Douglas J. Casa, Elaine C. Lee, Brendon P. McDermott, Jennifer F. Klau, Liliana Jimenez, Laurent Le Bellego, Emmanuel Chevillotte, and Harris R. Lieberman. Mild dehydration affects mood in healthy young women. *Journal of Nutrition*, 142(2):382–388, 2012. doi: 10.3945/jn.111.142000.

Athanasius of Alexandria. *On the Incarnation*. St Vladimir's Seminary Press, Yonkers, New York, 2011.

Deepika Bagga, Johanna Louise Reichert, Karl Koschutnig, Christoph Stefan Aigner, Peter Holzer, Kaisa Koskinen, Christine Moissl-Eichinger, and Veronika Schöpf. Probiotics drive gut microbiome triggering emotional brain signatures. *Gut Microbes*, 9(6):486–496, 2018. doi: 10.1080/19490976.2018.1460015.

David S. Black, Steve W. Cole, Michael R. Irwin, Elizabeth Breen, Natalie M. St. Cyr, Nora Nazarian, Dharma S. Khalsa, and Helen Lavretsky. Yogic meditation reverses NF-kB and IRF-related transcriptome dynamics in leukocytes of family dementia caregivers in a randomized controlled trial. *Psychoneuroendocrinology*, 38(3):348–355, 2013. doi: 10.1016/j.psyneuen.2012.06.011.

Judson A. Brewer, Patrick D. Worhunsky, Jeremy R. Gray, Yi-Yuan Tang, Jochen Weber, and Hedy Kober. Meditation experience is associated with differences in default mode network activity and connectivity. *Proceedings of the National Academy of Sciences*, 108 (50):20254–20259, 2011. doi: 10.1073/pnas.1112029108.

Jason M. Brill. A proposal for inferential evidence of the DNA phantom effect. *DNA Decipher Journal*, 2(2):125–135, 2012. URL https://dnadecipher.com/index.php/ddj/article/view/24. The author states that the original effect had not, to his knowledge, been successfully replicated.

Gudrun Bühnemann. The Saradatilakatantra on yoga: A new edition and translation of chapter 25. *Bulletin of the School of Oriental and African Studies*, 74(2):205–235, 2011. doi: 10.1017/S0041977X11000036.

Adenauer G. Casali, Olivia Gosseries, Mario Rosanova, Mélanie Boly, Simone Sarasso, Karina R. Casali, Silvia Casarotto, Marie-Aurélie Bruno, Steven Laureys, Giulio Tononi, and Marcello Massimini. A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, 5(198):198ra105, 2013. doi: 10.1126/scitranslmed.3006294.

Yuan Cha, Christopher J. Murray, and Judith P. Klinman. Hydrogen tunneling in enzyme reactions. *Science*, 243(4896):1325–1330, 1989. doi: 10.1126/science.2646716.

Sarah L. Chellappa, Christopher J. Morris, and Frank A. J. L. Scheer. Daily circadian misalignment impairs human cognitive performance task-dependently. *Scientific Reports*, 8(1):3041, 2018. doi: 10.1038/s41598-018-20707-4.

Brian P. Copenhaver. *Hermetica: The Greek Corpus Hermeticum and the Latin Asclepius in a New English Translation*. Cambridge University Press, Cambridge, 1992.

Paul Covington, Jay Adams, and Emre Sargin. Deep neural networks for YouTube recommendations. In *Proceedings of the 10th ACM Conference on Recommender Systems*, pages 191–198, 2016. doi: 10.1145/2959100.2959190.

Francis Crick. Central dogma of molecular biology. *Nature*, 227(5258):561–563, 1970. doi: 10.1038/227561a0.

Raúl de la Fuente-Fernández, Thomas J. Ruth, Vesna Sossi, Michael Schulzer, Donald B. Calne, and A. Jon Stoessl. Expectation and dopamine release: Mechanism of the placebo effect in parkinson's disease. *Science*, 293(5532):1164–1166, 2001. doi: 10.1126/science.1060937.

Jon G. Dean, Tiecheng Liu, Sean Huff, Ben Sheler, Steven A. Barker, Rick J. Strassman, Michael M. Wang, and Jimo Borjigin. Biosynthesis and extracellular concentrations of N,N-dimethyltryptamine (DMT) in mammalian brain. *Scientific Reports*, 9(1):9333, 2019. doi: 10.1038/s41598-019-45812-w.

Michel Desmurget, Karen T. Reilly, Nathalie Richard, Alexandru Szathmari, Carmine Mottolese, and Angela Sirigu. Movement intention after parietal cortex stimulation in humans. *Science*, 324(5928):811–813, 2009. doi: 10.1126/science.1169896.

Hong-Guang Duan, Valentyn I. Prokhorenko, Richard J. Cogdell, Khuram Ashraf, Amy L. Stevens, Michael Thorwart, and R. J. Dwayne Miller. Nature does not rely on long-lived electronic quantum coherence for photosynthetic energy transfer. *Proceedings of the National Academy of Sciences*, 114(32):8493–8498, 2017. doi: 10.1073/pnas.1702261114.

Fallon Durant, Johanna Bischof, Chris Fields, Junji Morokuma, Joshua LaPalme, Alison Hoi, and Michael Levin. The role of early bioelectric signals in the regeneration of planarian anterior/posterior polarity. *Biophysical Journal*, 116(5):948–961, 2019. doi: 10.1016/j.bpj.2019.01.029.

Matthew S. Ganio, Lawrence E. Armstrong, Douglas J. Casa, Brendon P. McDermott, Elaine C. Lee, Linda M. Yamamoto, Stefania Marzano, Rebecca M. Lopez, Liliana Jimenez, Laurent Le Bellego, Emmanuel Chevillotte, and Harris R. Lieberman. Mild dehydration impairs cognitive performance and mood of men. *British Journal of Nutrition*, 106(10):1535–1543, 2011. doi: 10.1017/S0007114511002005.

Peter P. Gariaev, Alexander A. Kokaya, Irina V. Mukhina, Elizaveta A. Leonova-Gariaeva, Nina G. Kokaya, Olga A. Davydova, Sergey V. Tertichniy, and Alexander V. Tovmash.

Principles of linguistic-wave genetics. *DNA Decipher Journal*, 1(1):11–24, 2011. URL https://dnadecipher.com/index.php/ddj/article/view/4.

Jan-Sebastian Grigoleit, Jennifer S. Kullmann, Oliver T. Wolf, Florian Hammes, Alexander Wegner, Stephanie Jablonowski, Harald Engler, Elke Gizewski, Reiner Oberbeck, and Manfred Schedlowski. Dose-dependent effects of endotoxin on neurobehavioral functions in humans. *PLoS ONE*, 6(12):e28330, 2011. doi: 10.1371/journal.pone.0028330.

Kevin D. Hall, Alexis Ayuketah, Robert Brychta, Hongyi Cai, Thomas Cassimatis, Kong Y. Chen, Stephanie T. Chung, Elise Costa, Amber Courville, Valerie Darcey, et al. Ultra-processed diets cause excess calorie intake and weight gain: An inpatient randomized controlled trial of ad libitum food intake. *Cell Metabolism*, 30(1):67–77.e3, 2019. doi: 10.1016/j.cmet.2019.05.008.

Stuart Hameroff and Roger Penrose. Consciousness in the universe: A review of the "Orch OR" theory. *Physics of Life Reviews*, 11(1):39–78, 2014. doi: 10.1016/j.plrev.2013.08.002.

Ian A. Hatton, Eric D. Galbraith, Nono S. C. Merleau, Teemu P. Miettinen, Benjamin McDonald Smith, and Jeffery A. Shander. The human cell count and size distribution. *Proceedings of the National Academy of Sciences*, 120(39):e2303077120, 2023. doi: 10.1073/pnas.2303077120.

Stephen B. Hladky and Margery A. Barrand. Mechanisms of fluid movement into, through and out of the brain: evaluation of the evidence. *Fluids and Barriers of the CNS*, 11(1): 26, 2014. doi: 10.1186/2045-8118-11-26.

International Commission on Non-Ionizing Radiation Protection. Guidelines for limiting exposure to electromagnetic fields (100 khz to 300 ghz). *Health Physics*, 118(5):483–524, 2020. doi: 10.1097/HP.0000000000001210.

C. G. Jung. *Psychology and Alchemy*, volume 12 of *Collected Works of C. G. Jung*. Princeton University Press, Princeton, 2 edition, 1968.

Eric R. Kandel, John D. Koester, Sarah H. Mack, and Steven A. Siegelbaum. *Principles of Neural Science*. McGraw Hill, New York, 6 edition, 2021. ISBN 9781259642234.

Michael Levin. Bioelectric signaling: Reprogrammable circuits underlying embryogenesis, regeneration, and cancer. *Cell*, 184(8):1971–1989, 2021. doi: 10.1016/j.cell.2021.02.034.

David E. Nichols. N,N-dimethyltryptamine and the pineal gland: Separating fact from myth. *Journal of Psychopharmacology*, 32(1):30–36, 2018. doi: 10.1177/0269881117736919.

Gitt Panitchayangkoon, Dugan Hayes, Kelly A. Fransted, Justin R. Caram, Elad Harel, Jianzhong Wen, Robert E. Blankenship, and Gregory S. Engel. Long-lived quantum coherence in photosynthetic complexes at physiological temperature. *Proceedings of the National Academy of Sciences*, 107(29):12766–12770, 2010. doi: 10.1073/pnas.1005484107.

Aristotle Papanikolaou. Theosis. In Bernard McGinn and John Strawn, editors, *The Oxford Handbook of Mystical Theology*, pages 568–585. Oxford University Press, Oxford, 2020. doi: 10.1093/oxfordhb/9780198722380.013.29.

Bradley J. Roth. The magnetocardiogram. *APL Bioengineering*, 8(2):021501, 2024. doi: 10.1063/5.0201950.

Britton A. Sauerbrei and J. Andrew Pruszynski. The brain works at more than 10 bits per second. *Nature Neuroscience*, 28(7):1365–1366, 2025. doi: 10.1038/s41593-025-01997-0.

Paschal Sheeran, Thomas L. Webb, and Peter M. Gollwitzer. The interplay between goal intentions and implementation intentions. *Personality and Social Psychology Bulletin*, 31 (1):87–98, 2005. doi: 10.1177/0146167204271308.

Cary Stothart, Ainsley Mitchum, and Courtney Yehnert. The attentional cost of receiving a cell phone notification. *Journal of Experimental Psychology: Human Perception and Performance*, 41(4):893–897, 2015. doi: 10.1037/xhp0000100.

Mkael Symmonds, Julian J. Emmanuel, Megan E. Drew, Rachel L. Batterham, and Raymond J. Dolan. Metabolic state alters economic decision making under risk in humans. *PLoS ONE*, 5(6):e11090, 2010. doi: 10.1371/journal.pone.0011090.

Max Tegmark. Importance of quantum decoherence in brain processes. *Physical Review E*, 61(4):4194–4206, 2000. doi: 10.1103/PhysRevE.61.4194.

Joshua D. Upshaw, Carl E. Stevens, Giorgio Ganis, and Darya L. Zabelina. The hidden cost of a smartphone: The effects of smartphone notifications on cognitive control from a behavioral and electrophysiological perspective. *PLoS ONE*, 17(11):e0277220, 2022. doi: 10.1371/journal.pone.0277220.

David R. Vago and David A. Silbersweig. Self-awareness, self-regulation, and self-transcendence (S-ART): a framework for understanding the neurobiological mechanisms of mindfulness. *Frontiers in Human Neuroscience*, 6:296, 2012. doi: 10.3389/fnhum.2012.00296.

Lev Vaidman. Many-worlds interpretation of quantum mechanics. The Stanford Encyclopedia of Philosophy, 2022. URL https://plato.stanford.edu/entries/qm-manyworlds/. Fall 2022 edition.

Samuel J. Virolainen, Andrew VonHandorf, Kenyatta C. M. F. Viel, Matthew T. Weirauch, and Leah C. Kottyan. Gene–environment interactions and their impact on human health. *Cellular and Molecular Life Sciences*, 80(1):1–11, 2023. doi: 10.1038/s41435-022-00192-6.

Tor D. Wager, James K. Rilling, Edward E. Smith, Alex Sokolik, Kenneth L. Casey, Richard J. Davidson, Stephen M. Kosslyn, Robert M. Rose, and Jonathan D. Cohen. Placebo-induced changes in fMRI in the anticipation and experience of pain. *Science*, 303(5661):1162–1167, 2004. doi: 10.1126/science.1093065.

David Gordon White. *The Alchemical Body: Siddha Traditions in Medieval India*. University of Chicago Press, Chicago, 1996.

Paul Whitney, John M. Hinson, Melinda L. Jackson, and Hans P. A. Van Dongen. Feedback blunting: Total sleep deprivation impairs decision making that requires updating based on feedback. *Sleep*, 38(5):745–754, 2015. doi: 10.5665/sleep.4668.

Wendy Wood, Leona Tam, and Melissa Guerrero Witt. Changing circumstances, disrupting habits. *Journal of Personality and Social Psychology*, 88(6):918–933, 2005. doi: 10.1037/0022-3514.88.6.918.

Marjorie H. Woollacott, Yvonne Kason, and Russell D. Park. Investigation of the phenomenology, physiology and impact of spiritually transformative experiences: kundalini awakening. *Explore*, 17(6):525–534, 2021. doi: 10.1016/j.explore.2020.07.005.

David Bryce Yaden, Jonathan Haidt, Ralph W. Hood, David R. Vago, and Andrew B. Newberg. The varieties of self-transcendent experience. *Review of General Psychology*, 21 (2):143–160, 2017. doi: 10.1037/gpr0000102.

Wan X. Yao, Vinoth K. Ranganathan, Didier Allexandre, Vlodek Siemionow, and Guang H. Yue. Kinesthetic imagery training of forceful muscle contractions increases brain signal and muscle strength. *Frontiers in Human Neuroscience*, 7:561, 2013. doi: 10.3389/fnhum.2013.00561.

Jieyu Zheng and Markus Meister. The unbearable slowness of being: Why do we live at 10 bits/s? *Neuron*, 113(2):192–204, 2025. doi: 10.1016/j.neuron.2024.11.008.

Theodore J. Zwang, Edmund C. M. Tse, and Jacqueline K. Barton. Sensing DNA through DNA charge transport. *ACS Chemical Biology*, 13(7):1799–1809, 2018. doi: 10.1021/acschembio.8b00347.